Linear Regression

An ice cream shop recorded its sales on various days of the year, together with the weather on that day, to try to determine if there is a meaningful relationship between temperature and sales. The observations are plotted below, in blue. The red line is an example of a simple linear regression obtained by minimizing the total squared error (each one is shown as a red box) from the points to the line.

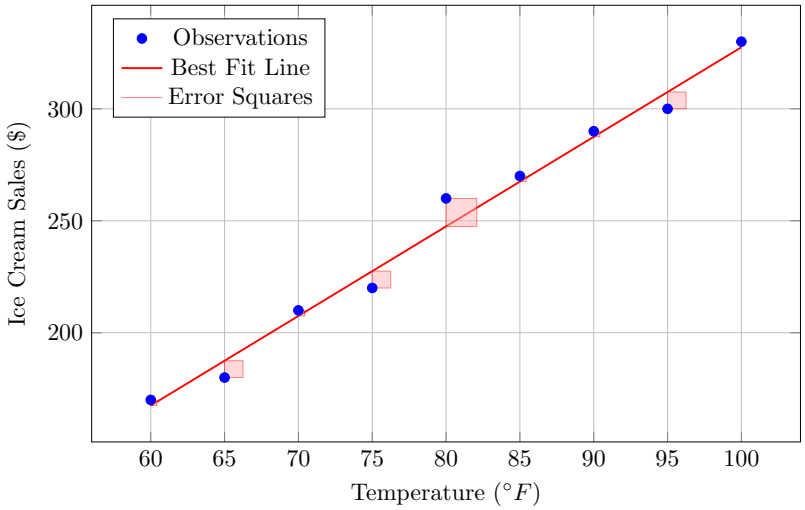


Figure 1: Temperature vs. Ice Cream Sales with the Best Fit Line

1 Ordinary Least-Squares

Imagine we are given a data set $\mathcal{D} = \{(\boldsymbol{x}_i, y_i) \mid \boldsymbol{x}_i \in \mathbb{R}^d, y_i \in \mathbb{R}, i \in \{1, \ldots, m\}\}$ of size m, where each observation $\boldsymbol{x}_i \in \mathbb{R}^d$ is d-dimensional, meaning each observation is described by d **features**.

In a regression model, when we "regress Y on X," Y is called the **response**, or **dependent**, variable, while the features are called the **explanatory** variables, or the **predictors**.

The feature values, which are presented as tabular data, can be encoded in a matrix $X \in \mathbb{R}^{m \times d}$, as follows:

$$X = \begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_m \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \ldots & x_{1d} \\ x_{21} & x_{22} & \ldots & x_{2d} \\ \vdots & & & \\ x_{m1} & x_{m2} & \ldots & x_{md} \end{bmatrix}$$

Each $\boldsymbol{x}_i$ in X is a row vector, i.e., an element of $\mathbb{R}^d$. When the number of features $d = 1$, the problem is called **simple regression**. Otherwise, it is called **multiple regression**.

[1] This notes were compiled in conjunction with Professor Eric Ewing.

The response variables $\boldsymbol{y} \in \mathbb{R}^m$ are given by a column vector:

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

The goal is to find a set of **weights** $\boldsymbol{w} \in \mathbb{R}^d$, or **coefficients**, s.t. the model's **estimates**,[2] or **predictions**,[3] $X\boldsymbol{w}$, approximate $\boldsymbol{y}$:

$$X\boldsymbol{w} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1d} \\ x_{21} & x_{22} & \dots & x_{2d} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{md} \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} w_1 x_{11} + w_2 x_{12} + \ldots + w_d x_{1d} \\ w_2 x_{21} + w_2 x_{12} + \ldots + w_d x_{1d} \\ \vdots \\ w_1 x_{m1} + w_2 x_{m2} + \ldots + w_d x_{md} \end{bmatrix} \approx \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \boldsymbol{y}$$

Observe that $X\boldsymbol{w} = \boldsymbol{y}$ is a system of m equations with d unknowns. A system of equations can be solved when it has the same number of equations and unknowns, i.e., when $n = d$. If $n < d$, i.e., if the matrix is wider than it is tall, then the system is **underdetermined**; it has infinitely many solutions. If $n > d$, i.e., if the matrix is taller than it is wide, which is usually the case in regression, the system is **overdetermined**, which means it has no solution. Consequently, rather than aim to solve this system of equations exactly, we solve for weights that minimize the difference between the predicted values $X\boldsymbol{w}$ and the observed values y.

Define the **residual** of a candidate solution $\boldsymbol{w}$ as the difference between $\boldsymbol{y}$ and $X\boldsymbol{w}$, i.e.,

$$\text{residual}(\boldsymbol{w}) = \boldsymbol{y} - X\boldsymbol{w}$$

The **least squares objective** is to minimize the *square* of this residual value: $\text{loss}(\boldsymbol{w}) \doteq \text{residual}^2(\boldsymbol{w})$.

Why square the residuals? "My reasons are better than they seem."[4]

1. Squaring errors precludes positive and negative errors from canceling one another out (and so does using absolute error, based on absolute values)
2. Squaring errors penalizes larger errors far more than smaller ones, thereby forcing the model to prioritize minimizing large errors over smaller ones
3. The ensuing estimator matches that which is derived by the maximum likelihood principle, assuming the data are normally distributed
4. This form of regression affords a closed-form solution

As $\text{residual}(\boldsymbol{w})$ is an m-dimensional column vector, we square it as follows:

$$\text{loss}(\boldsymbol{w}) = \text{residual}^2(\boldsymbol{w}) = (\boldsymbol{y} - X\boldsymbol{w})^T(\boldsymbol{y} - X\boldsymbol{w}) = \sum_{i=1}^{m} (\boldsymbol{y} - X\boldsymbol{w})_i^2$$

In other words, we are interested in minimizing the sum of the squared residuals.

As usual, to minimize an objective, we take its derivative and set it equal to zero.

[2] statistical nomenclature
[3] machine learning nomenclature
[4] A modern English translation of William Shakespeare, *All's Well that Ends Well*

First, let's simplify the loss:

$$\begin{aligned}
&\text{loss}(\boldsymbol{w}) && (1)\\
&= (\boldsymbol{y} - X\boldsymbol{w})^T(\boldsymbol{y} - X\boldsymbol{w}) && (2)\\
&= \boldsymbol{y}^T\boldsymbol{y} - \boldsymbol{y}^T X\boldsymbol{w} - (X\boldsymbol{w})^T\boldsymbol{y} + (X\boldsymbol{w})^T X\boldsymbol{w} && (3)\\
&= \boldsymbol{y}^T\boldsymbol{y} - \boldsymbol{y}^T X\boldsymbol{w} - \boldsymbol{w}^T X^T\boldsymbol{y} + \boldsymbol{w}^T X^T X\boldsymbol{w} && (4)\\
&= \boldsymbol{y}^T\boldsymbol{y} - (\boldsymbol{y}^T X\boldsymbol{w})^T - \boldsymbol{w}^T X^T\boldsymbol{y} + \boldsymbol{w}^T X^T X\boldsymbol{w} && (5)\\
&= \boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{w}^T X^T\boldsymbol{y} + \boldsymbol{w}^T X^T X\boldsymbol{w} && (6)
\end{aligned}$$

In Equations 3 and 6, we apply the identity $(AB)^T = B^T A^T$, while Equation 5 follows from the fact that $\boldsymbol{y}^T X\boldsymbol{w}$ is a scalar, so that its value equals its transpose.

Now let's take the derivative:

$$\begin{aligned}
&\nabla_{\boldsymbol{w}}\text{loss}(\boldsymbol{w}) && (7)\\
&= \nabla_{\boldsymbol{w}}\boldsymbol{y}^T\boldsymbol{y} - 2\nabla_{\boldsymbol{w}}\boldsymbol{w}^T X^T\boldsymbol{y} + \nabla_{\boldsymbol{w}}\boldsymbol{w}^T X^T X\boldsymbol{w} && (8)\\
&= -2X^T y + 2X^T Xw && (9)
\end{aligned}$$

Equation 9 relies on the following fact: $\nabla_{\boldsymbol{x}}\boldsymbol{x}^T A\boldsymbol{x} = 2A\boldsymbol{x}$.

Setting this derivative equal to zero yields a **closed-form** (i.e., analytical) solution to linear regression:

$$\begin{aligned}
-2X^T y + 2X^T Xw &= 0 && (10)\\
X^T Xw &= X^T y && (11)\\
w &= \underbrace{(X^T X)^{-1}X^T}_{\text{pseudo-inverse}} y && (12)
\end{aligned}$$

The matrix $X^T X$ might not be invertible. This matrix is only invertible when the columns of X are linearly independent, so that they span $\mathbb{R}^d$. The pseudo-inverse $(X^T X)^{-1}X^T$, however, always exists, and can be calculated via a **singular value decomposition**.

We have thus derived a $\boldsymbol{w}$ that satisfies the **first-order optimality condition**, namely $\nabla_{\boldsymbol{w}}\text{loss}(\boldsymbol{w}) = 0$. To conclude that $\boldsymbol{w}$ is a minimum, we also need to establish that it satisfies the **second-order optimality condition**, namely that $\nabla^2_{\boldsymbol{w}}\text{loss}(\boldsymbol{w}) \geq 0$. Indeed, $\nabla^2_{\boldsymbol{w}}\text{loss}(\boldsymbol{w}) = 2X^T X \geq 0$.

Alternative Derivation We describe an alternative shorter, geometric, and arguably more intuitive way, to derive the least-squares estimator. The only way the residual $\boldsymbol{y} - X\boldsymbol{w}$ can be zero is if $\boldsymbol{y}$ is a **linear combination** of the columns of X: i.e., if $\boldsymbol{y}$ lies in the **column span** of X. As most of the time it does not, our stated goal of minimizing the residual (equivalently, the total value of the squared residuals) can instead be understood as minimizing the distance from the residual to the column span of X. To minimize this distance, the residual should be **projected** onto the column span of X; that is, it should be **orthogonal** to each and every column of X: for all $j \in \{1, \ldots, n\}$, i.e., $(\boldsymbol{y} - X\boldsymbol{w}) \cdot X_j = \boldsymbol{0}_d$. Equivalently, but expressed more compactly in matrix notation, $(\boldsymbol{y} - X\boldsymbol{w})^T X = \boldsymbol{0}_d$. This requirement implies:

$$\begin{aligned}
X^T(\boldsymbol{y} - X\boldsymbol{w}) &= \boldsymbol{0}_d && (13)\\
X^T X\boldsymbol{w} &= X^T\boldsymbol{y} && (14)\\
\boldsymbol{w} &= (X^T X)^{-1}X^T\boldsymbol{y} && (15)
\end{aligned}$$

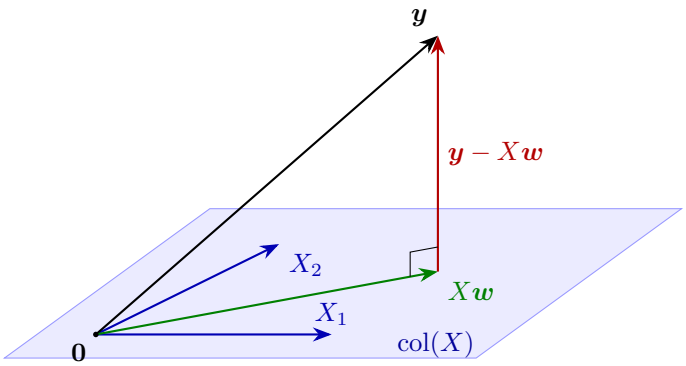


Figure 2: The vector $\boldsymbol{y}$ generally lies outside the column span of X. The closest point $X\boldsymbol{w}$ in $\mathrm{col}(X)$ is the orthogonal projection of $\boldsymbol{y}$, which makes the residual $\boldsymbol{y} - X\boldsymbol{w}$ orthogonal to every column X_j of X.

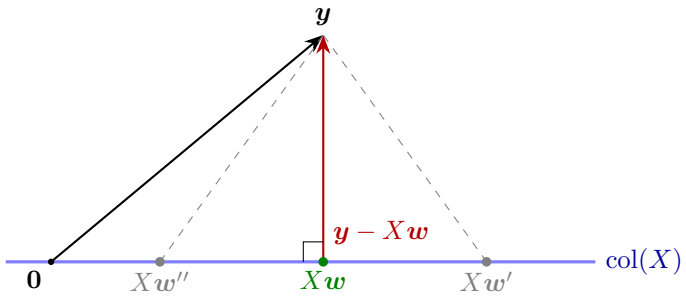


Figure 3: Among all points $X\boldsymbol{w}$ in $\mathrm{col}(X)$, the one whose residual is orthogonal to $\mathrm{col}(X)$ is closest to $\boldsymbol{y}$: any other choice ($X\boldsymbol{w}'$, $X\boldsymbol{w}''$) yields a longer residual, by the Pythagorean theorem.